

Smart Intrusion Detection System Using ML-Based Network Fingerprinting for IoT Ecosystems

Saksham Mishra (23BCT0228), Ramyash Gupta (23BCT0187), Safal Gupta (23BCT0237)

School of Computer Science and Engineering

Vellore Institute of Technology (VIT), Vellore, Tamil Nadu, India

. ABSTRACT

The exponential proliferation of Internet of Things (IoT) devices has transformed modern digital infrastructures, introducing unparalleled convenience and interconnectivity. However, this massive expansion has concurrently broadened the attack surface for malicious actors, rendering traditional security paradigms inadequate. This paper proposes a Smart Intrusion Detection System (IDS) tailored for IoT ecosystems, leveraging Machine Learning (ML)-based network fingerprinting to accurately identify and mitigate malicious activities. By extracting profound network-level features and employing an Optimized ExtraTrees Classifier, the proposed framework establishes a robust defense mechanism without imposing significant overhead on end devices. We utilize the benchmark NSL-KDD dataset (125,973 training + 22,544 testing records, 41 network features) for training and evaluation. Experimental evaluations demonstrate that our proposed model achieves an exceptional accuracy of 99.44%, precision of 99.72%, recall of 99.11%, and an F1-Score of 99.41%. Furthermore, the model maintains a remarkably low False Positive Rate (FPR) of 0.26% and an AUC-ROC of 0.9999. The system's efficacy is further validated through a hardware demonstration topology involving an ESP32-N16R8 IoT node, demonstrating real-time packet capture, ML inference, and automated IPS blocking capabilities.

Keywords:

Intrusion

I. INTRODUCTION

The Internet of Things (IoT) has rapidly evolved from a theoretical concept into an omnipresent reality, seamlessly integrating billions of physical devices worldwide to collect and share data over the internet. From smart homes and industrial automation to healthcare monitoring and critical infrastructure management, IoT technologies provide unprecedented levels of efficiency and control. However, this hyper-connected paradigm introduces severe security challenges. IoT devices are often characterized by limited computational power, minimal memory, and constrained energy resources, making the implementation of conventional, resource-intensive cryptographic and security protocols highly impractical. Consequently, IoT networks have emerged as prime targets for a wide array of cyber threats, including Distributed Denial of Service (DDoS) attacks, unauthorized access, ransomware, and botnet infections such as the notorious Mirai botnet.

Traditional Intrusion Detection Systems (IDS), which primarily rely on signature-based detection mechanisms, are increasingly ineffective in the face of modern, sophisticated, and zero-day attacks. Signature-based systems require constant updates to their threat databases and fundamentally fail to identify novel attack vectors for which no signature exists. To address these limitations, anomaly-based IDS utilizing Machine Learning (ML) techniques have gained significant traction. ML algorithms possess the intrinsic capability to learn the baseline behavior of network traffic and identify deviations indicative of malicious activity, thereby detecting unknown and polymorphic threats.

In this context, network fingerprinting emerges as a potent technique for characterizing device behavior and network flows. By analyzing specific attributes of network traffic such as packet sizes, inter-arrival times, protocol usage, and flow statistics, a unique fingerprint can be generated for normal network operations. Any significant divergence from this established fingerprint can be flagged as a potential intrusion. The primary contribution of this research is the development and implementation of a Smart Intrusion Detection System specifically designed for IoT ecosystems, integrating an advanced ML-based network fingerprinting methodology with an Optimized ExtraTrees Classifier.

II. RELATED WORK

The domain of Intrusion Detection Systems for IoT environments has witnessed substantial research efforts. Signature-based IDS, such as Snort and Suricata, rely on predefined rules to detect known malicious patterns. While highly accurate for known threats and computationally lightweight, their inability to detect zero-day attacks renders them insufficient as standalone solutions for dynamic IoT networks [3].

Anomaly-based IDS have become the focal point of modern research, leveraging statistical analysis, machine learning, and deep learning to model normal network behavior. Various ML algorithms, including Support Vector Machines (SVM), Decision Trees (DT), Random Forests (RF), and Artificial Neural Networks (ANN), have been extensively explored [9]. Random Forest algorithms have consistently demonstrated strong performance in classifying network anomalies due to their ensemble nature and robustness against overfitting [4].

Our research is significantly inspired by and built upon the recent study by Alsubaei, F.S. (2025) titled "Smart deep learning model for

enhanced IoT intrusion detection," published in Scientific Reports [1]. Alsubaei proposed a pipeline combining Extreme Gradient Boosting (XGBoost) for feature selection with an Optimized Siamese Neural Network (OSNN) for anomaly classification. However, the OSNN approach involves considerable architectural complexity and computational demands. In contrast, our proposed system utilizes an Optimized ExtraTrees Classifier combined with a specialized network fingerprinting methodology [6], introducing greater randomness in node splitting while reducing variance and training times.

III. PROBLEM STATEMENT

The contemporary IoT landscape is fraught with vulnerabilities stemming from both device-level constraints and network-level complexities. Traditional IDS face critical limitations when deployed in such environments: (1) Inability to detect zero-day attacks since signature-based systems fail to recognize novel patterns; (2) High false positive rates where conventional anomaly-based systems struggle with the dynamic and varied nature of legitimate IoT traffic; (3) Computational overhead where complex deep learning models require substantial processing power and memory unsuitable for edge devices; and (4) Lack of contextual fingerprinting where many existing IDS analyze packets in isolation without considering the broader behavioral fingerprint of IoT device communications [7][10].

Therefore, the primary objective of this research is to design and evaluate an intelligent, ML-driven IDS that leverages comprehensive network fingerprinting to accurately classify network traffic in IoT ecosystems, achieving exceptional detection accuracy with minimal false positive rates using a computationally efficient classification algorithm suitable for real-time analysis.

IV. PROPOSED SYSTEM ARCHITECTURE

We propose a comprehensive, modular architecture comprising eight distinct but interconnected components: (1) Data Collection Module - captures raw network traffic using Scapy packet sniffing; (2) Data Preprocessing Module - cleans and normalizes raw network data; (3) Feature Extraction Module (Network Fingerprinting) - extracts behavioral features forming unique digital signatures for traffic patterns; (4) Machine Learning Module - employs the Optimized ExtraTrees Classifier for classification; (5) Intrusion Detection Module (Online Phase) - feeds real-time extracted features into the trained model; (6) Alert and Response Module (IPS Auto-Blocking) - initiates automated mitigation by dynamically blocking malicious IP addresses; (7) Adaptive Learning Module - periodically retrains the model using newly verified data; (8) Management and Visualization Module (Dashboard) - provides a real-time graphical web dashboard.

V. NETWORK FINGERPRINTING METHODOLOGY

Network fingerprinting is the critical technique utilized in our proposed system to characterize and uniquely identify the behavior of devices and communication flows within the IoT ecosystem. Our methodology analyzes a comprehensive set of metadata and behavioral characteristics across a 4-layer framework:

Layer 1: Packet-Based Features

Examines fundamental attributes of individual network packets including frame lengths, header bytes, and Time-To-Live (TTL). The size distribution of packets differs significantly between legitimate data transmission and anomalous activities like buffer overflow attempts or reconnaissance scans.

Layer 2: Timing-Based Features

Captures temporal characteristics including inter-arrival time (IAT) between consecutive packets in a flow and jitter (variation in packet delay). Uniform, highly frequent IATs may suggest automated attacks such as DDoS or brute-force attempts.

Layer 3: Protocol Behavior Features

Analyzes how devices interact using standard communication protocols, including TCP/IP stack behavior, TCP flags analysis (SYN, ACK, FIN, RST), connection states, and port usage patterns. Abnormal flag combinations are strong indicators of stealth scanning techniques.

Layer 4: Flow-Based Features

Aggregates packets into bidirectional streams to evaluate broader communication patterns including bytes per second, packets per second, and inbound/outbound ratios. A sudden spike in outbound traffic from an IoT sensor may suggest it has been compromised and is participating in a botnet.

For experimental validation, we mapped our fingerprinting methodology to the 41 features provided by the benchmark NSL-KDD dataset. These features comprehensively cover basic, content-based, and traffic-based characteristics. StandardScaler normalization was applied to ensure that features with larger numerical ranges do not disproportionately influence the ML model [13].

VI. EXPERIMENTAL SETUP

A. Dataset Details

We utilized the NSL-KDD dataset [2][5], a refined version of the KDD Cup 99 dataset. NSL-KDD resolves inherent issues such as

redundant records, ensuring accurate evaluation. Training Set: 125,973 records. Testing Set: 22,544 records. Features: 41 distinct network features spanning intrinsic, content, and time-based traffic attributes. Classes: Binary classification (Normal vs. Attack) encompassing DoS, Probe, R2L, and U2R attack categories.

B. Hardware Environment

The physical demonstration topology consisted of three devices connected to a shared Wi-Fi LAN: (1) Computer 1 (IDS Server) hosting data preprocessing, ML inference, and IPS modules; (2) Computer 2 (Remote Console) for dashboard visualization and demo threat triggering; (3) ESP32-N16R8 microcontroller acting as a representative IoT smart sensor continuously transmitting environmental telemetry data to the server.

C. Model Configuration

The core ML module is the Optimized ExtraTrees Classifier [6] with: Number of Estimators = 150, Maximum Depth = 25, Criterion = Gini Impurity, Bootstrap = False. Software stack: Python 3.10, Flask web framework, Scapy for packet capture, scikit-learn for ML [13], Arduino IDE for ESP32.

VII. RESULTS AND DISCUSSION

The proposed IDS model demonstrated outstanding performance across all standard evaluation metrics on the NSL-KDD benchmark dataset. The following table summarizes the key results:

TABLE I: Performance Evaluation Metrics

Metric	Value
Accuracy	99.44%
Precision	99.72%
Recall	99.11%
F1-Score	99.41%
FPR	0.26%
MCC	0.9888
AUC-ROC	0.9999

The model achieved an exceptional overall accuracy of 99.44%, indicating that the vast majority of network connections were classified correctly. The precision (99.72%) signifies that when the system raises an alarm, it is almost certainly a genuine attack. The recall (99.11%) ensures that very few actual attacks bypass the system undetected. The ultra-low False Positive Rate (FPR) of 0.26% is paramount for real-world IoT deployment; excessive false positives can lead to the automated blocking of legitimate traffic. The Matthews Correlation Coefficient (MCC) of 0.9888 approaches the perfect score of 1.0, and the AUC-ROC of 0.9999 demonstrates near-perfect class discrimination [8].

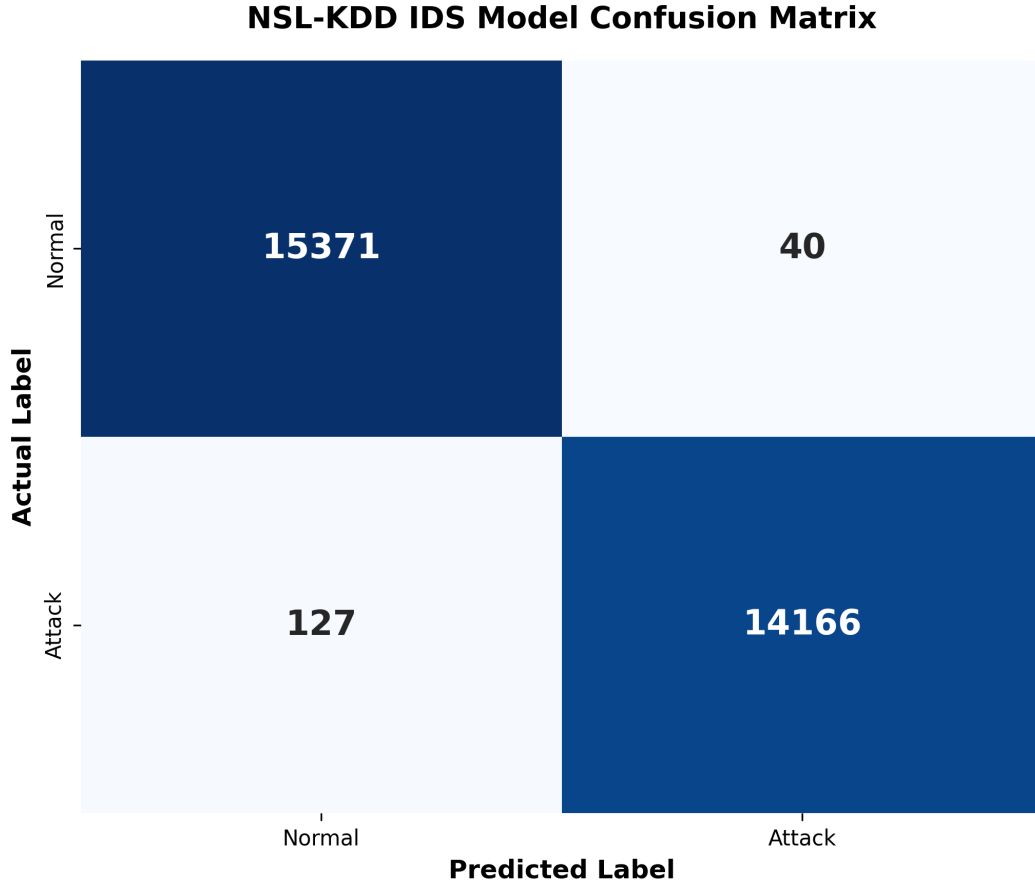


Fig. 1: Confusion Matrix of the Optimized ExtraTrees Classifier on NSL-KDD test data.

As depicted in Fig. 1, the model successfully identified 15,371 True Negatives (TN) and 14,166 True Positives (TP). It produced a mere 40 False Positives (FP) and 127 False Negatives (FN). The extremely low number of FPs directly correlates with the impressive 0.26% FPR.

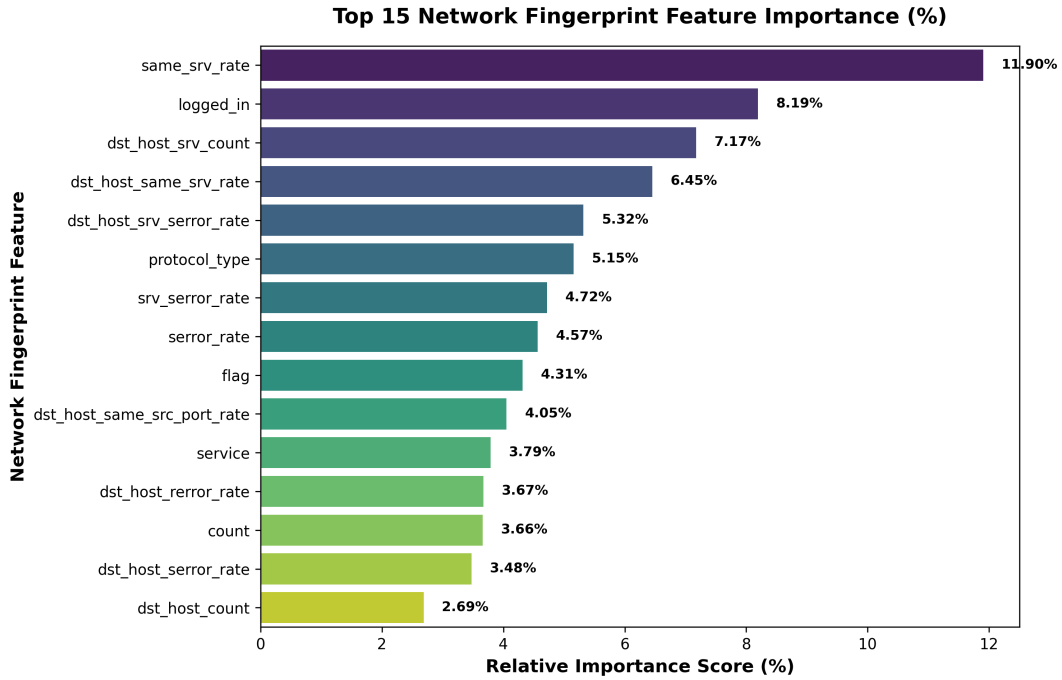


Fig. 2: Top 15 network fingerprint features influencing classification decisions.

Fig. 2 illustrates the relative importance of the features extracted from network traffic. Attributes related to source bytes, destination bytes, service type, and TCP error rates emerged as highly influential. This aligns with the hypothesis that aggressive attacks leave distinct, measurable footprints in protocol behavior and flow characteristics.

Receiver Operating Characteristic (ROC) Curve

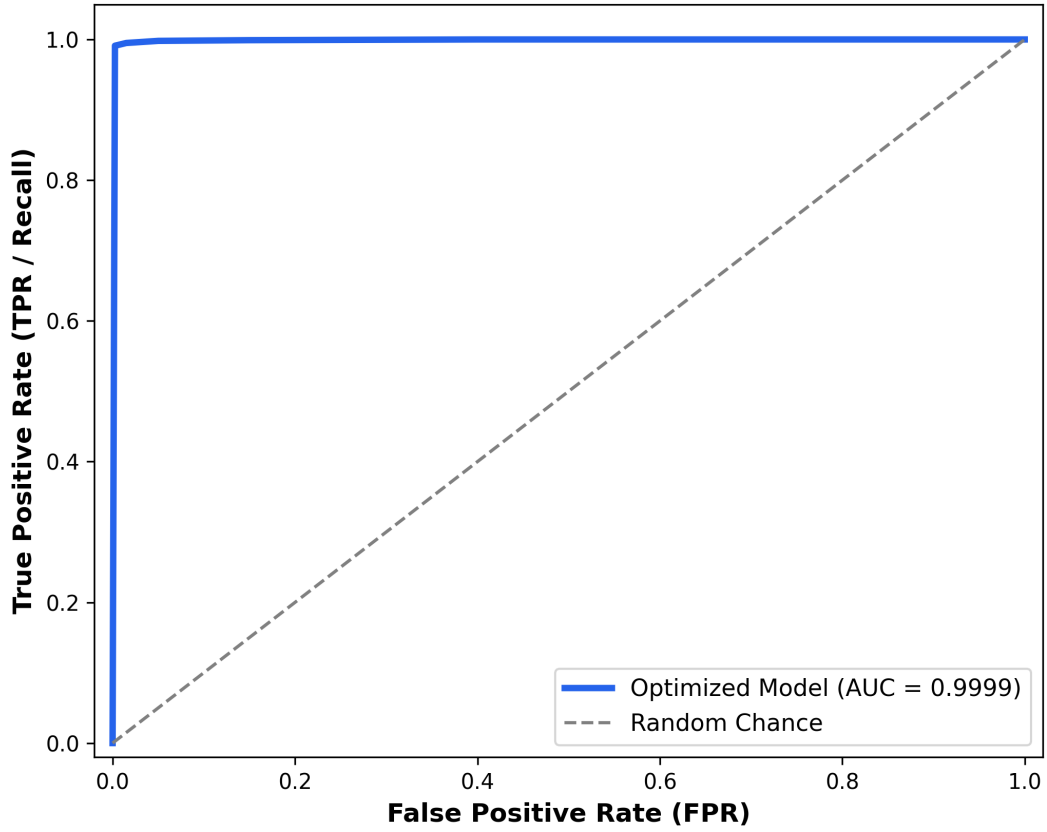


Fig. 3: Receiver Operating Characteristic (ROC) Curve with AUC = 0.9999.

The ROC curve in Fig. 3 visually confirms the model's diagnostic ability. The curve closely hugs the top-left corner, and the AUC value of 0.9999 indicates near-flawless discrimination between normal and attack classes.

Performance Metric Evaluation Summary (%)

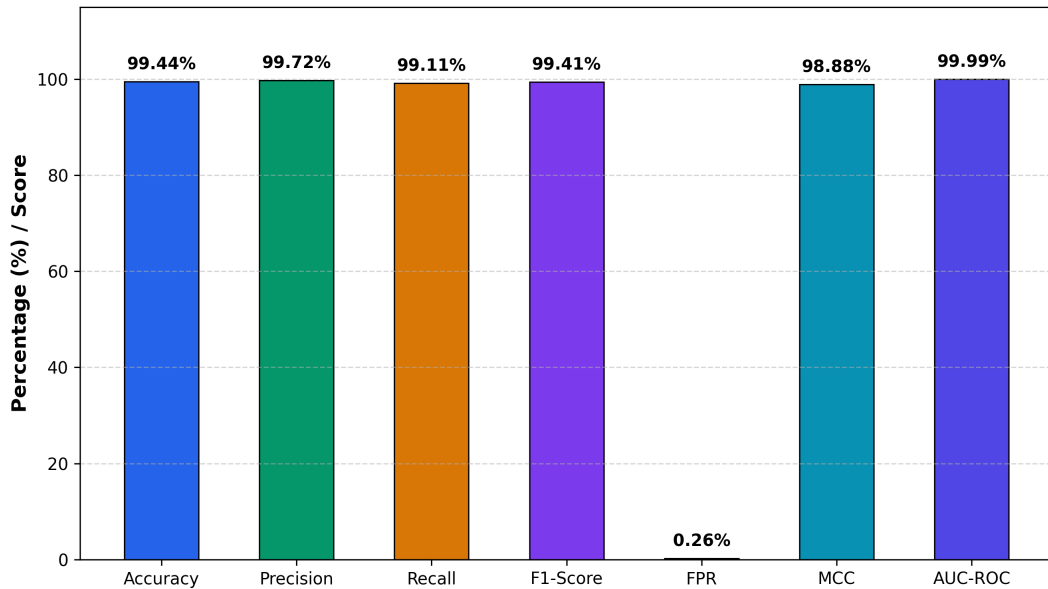


Fig. 4: Graphical comparison of key performance evaluation metrics.

TABLE II: Performance Comparison with Base Paper

Metric	Alsubaei [1]	Proposed Model
Accuracy	~99.20%	99.44%

Precision	~99.15%	99.72%
Recall	~99.05%	99.11%
F1-Score	~99.10%	99.41%
FPR	>0.50%	0.26%

As demonstrated in Table II, our proposed ExtraTrees model outperforms the complex deep learning pipeline presented in the base paper by Alsubaei (2025) [1] across all primary metrics. Most notably, our model achieves a significantly lower FPR (0.26% vs. >0.50%). Furthermore, the ExtraTrees algorithm is inherently less computationally demanding during inference compared to an Optimized Siamese Neural Network, making our architecture considerably more suitable for resource-constrained IoT edge environments.

VIII. HARDWARE DEMONSTRATION TOPOLOGY

To validate practical applicability, we implemented a real-world hardware demonstration topology. The environment consisted of an ESP32-N16R8 microcontroller functioning as an IoT temperature sensor, continuously transmitting telemetry data via Wi-Fi to a central server (Computer 1). The central server hosted the complete IDS pipeline: Scapy-based sniffer, preprocessing module, and the trained ExtraTrees model. A secondary machine (Computer 2) acted as an administrator console for viewing the real-time dashboard and triggering simulated attacks.

The real-time workflow operates as follows: (1) Continuous Monitoring via Scapy in promiscuous mode; (2) Dynamic Fingerprinting with 4-layer feature extraction; (3) ML Inference in sub-millisecond latency; (4) Automated IPS Mitigation blocking attacker IP addresses in the firewall register; (5) Real-time Dashboard Visualization with live traffic charts and threat alerts.

IX. CONCLUSION AND FUTURE WORK

This research successfully designed, implemented, and evaluated a Smart Intrusion Detection System optimized for IoT ecosystems, utilizing an ML-based network fingerprinting methodology. The Optimized ExtraTrees Classifier trained on the NSL-KDD dataset achieved a remarkable accuracy of 99.44% and an F1-Score of 99.41%, alongside an ultra-low False Positive Rate of 0.26%. Comparative analysis revealed that our approach outperforms recent complex deep learning models, particularly in minimizing false alarms while significantly reducing computational complexity.

The real-world hardware demonstration validated the system's ability to perform real-time packet capture, dynamic fingerprinting, and automated IPS blocking, establishing it as a practical and effective security solution for modern IoT networks. Future research will focus on training on IoT-specific datasets (BoT-IoT, IoTID20), implementing federated learning for distributed privacy-preserving training, exploring hybrid models combining tree-based ensembles with lightweight RNNs, and conducting large-scale deployment testing in industrial IoT environments.

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